

C & C++ programming

Handwritten Theory Notes

WEEK 1 MODULE - C & C++ PROGRAMMING

1. SYNTAX

C Programming

- C is a general purpose programming language.
- Developed by Dennis Ritchie in 1972.
- It is used for system programming, embedded systems and application development.
- Structure of C program:

```
#include <stdio.h>
int main()
{
    printf("Hello World!");
    return 0;
}
```

Basic Syntax in C:

- | | |
|----------------|--|
| 1. #include <> | → Preprocessor directive (header file) |
| 2. int main() | → Main function (program start) |
| 3. printf() | → Output function. |
| 4. return 0; | → End of program |
| 5. {} | → Block of code |
| 6. ; | → End of statement. |

• C++ Programming

- C++ is an extension of C.
- Developed by Bjarne Stroustrup.
- It supports Object-Oriented Programming (OOP).
- Structure of C++ Program.

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Hello world!";
    return 0;
}
```

• Basic Syntax in C++ :

1. #include <iostream>

→ Input/output library

2. using namespace std;

→ Uses standard namespace

3. cout <<

→ Output statement

4. cin >>

→ Input statement

5. ;

→ End of statement

6. {}

→ Block of code.

7. return 0

→ End of Program.

2. CONDITIONS AND LOOPS

A. CONDITIONS

- These are used to make decisions in a program. They execute different statements based on whether a condition is true or false.

1. if Statement

executes a block of code when the given statement is true.

• Syntax :

```
if (condition)
{
    // statements
}
```

• Example (C) :

```
int a=10;
if (a>5)
{
    printf("greater");
}
```

• Example (C++) :

```
int a=10;
if(a>5)
{
    cout << "greater";
}
```

2. if-else Statement

executes one block when condition is true and another block when condition is false.

• syntax :

```
if (condition)
{
    // statements
} else
{
    // statements.
}
```


• Example (C) :

```
int a=4;
if (a%2==0)
{
    printf("Even");
}
else
{
    printf("odd");
}
```

• Example (C++) :

```
int a=4;
if (a%2==0)
{
    cout << "Even";
}
else
{
    cout << "odd";
}
```

3. Nested if statement

→ An if statement placed inside another if statement.

• syntax:

```
if (condition1)
{
    if (condition2)
    {
        // statements
    }
}
```

• Ex (C) :

```
int a=10;
if (a>0)
{
    if (a%2==0)
        printf("Pos. Even")
    else
        printf("Pos. odd");
}
```

4. Switch statement

To select one option from multiple choices.

• Syntax:

```
switch (expression)
{
    case 1:
        // statements
        break;
    case 2:
        // statements
        break;
    ...
    default:
        // statements.
```

• Example (C) :

```
int day = 2;
switch (day)
{
    case 1: printf("Mon");
            break;
    case 2: printf("Tue");
            break;
    case 3: printf("Wed");
            break;
    default: printf("Invalid");
}
```

• Example (C++) :

```
int day = 2;
switch (day)
{
    case 1: cout << "Mon";
            break;
    case 2: cout << "Tue";
            break;
    case 3: cout << "Wed";
            break;
    default: cout << "Invalid";
}
```


8. Loops

Loops are used to execute a set of statements repeatedly until a specified condition becomes false.

1. for Loop

This is used when the no. of iterations is known.

Syntax:

```
for (initialisation; condition;
    increment/decrement)
{
    // statements
}
```

Example (C):

```
for (int i=1; i<=5; i++)
{
    printf ("%d", i);
}
```

Example (C++):

```
for (int i=1; i<=5; i++)
{
    cout << i << " ";
}
```

2. while Loop

The while loop executes as long as the condition remains true.

Syntax:

```
while (condition).
{
    // statements
}
```


Example (C) :

```
int i=1;
while (i<=5)
{
    printf("%d", i);
    i++;
}
```

Example (C++) :

```
int i=1;
while i<=5
{
    cout << i << " ";
    i++;
}
```

3. do-while loop

The do-while loop executes at least one time before checking the condition.

Syntax :

```
do
{
    // statements
}
while (condition);
```

Example (C) :

```
int i=1;
do
{
    printf("%d", i);
    i++;
}
while (i<=5);
```

Handwritten Codes

HANDWRITTEN CODE

A. C PROGRAM

1. Square

```
#include <stdio.h>

int main() {
    int i, k;
    for(i=0; i<5; i++)
    {
        for(k=0; k<5; k++)
        {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

output:

*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*

2. Hollow Square

```
#include <stdio.h>

int main() { int i, k;
    for(i=0; i<5; i++)
    {
        for(k=0; k<5; k++)
        {
            if(i==0 || i==4 || k==0 || k==4)
            {
                printf("* ");
            }
            else {
                printf(" ");
            }
        }
    }
}
```



```

    }
}
printf("\n");
}
return 0;
}

```

output -

```

* * * * *
*           *
*           *
*           *
* * * * *

```

3. Triangle

#include <stdio.h>

int main() {

int i, k;

for(i=0; i<5; i++) {

for(k=0; k<=i; k++) {

printf(" * ");

}

printf("\n");

}

return 0;

}

output -

```

*
* *
* * *
* * * *
* * * * *

```

4. Inverted Triangle -

#include <stdio.h>

int main() {

int i, k;

for(i=5; i>0; i--)

{

for(k=0; k<i; k++) {

printf(" * ");

}

printf("\n");

}

return 0.

}

output -

```

* * * * *
* * * *
* * *
* *
*

```


4. Pyramid

#include <stdio.h>

int main() {

int i, j, k;

for (i=0; i<5; i++) {

for (j=0; j<5-i; j++) {

printf(" ");

}

for (k=0; k<2*i-1; k++) {

printf(" * ");

}

printf("\n");

}

return 0;

}

output:

```

      *
     * * *
    * * * * *
   * * * * * *
  * * * * * * *
 * * * * * * *

```

5. Inverted Pyramid

#include <stdio.h>

int main() {

int i, j, k;

for (i=5; i>0; i--) {

for (j=0; j<5-i; j++) {

printf(" ");

}

for (k=0; k<2*i-1; k++) {

printf(" * ");

}

printf("\n");

}

return 0;

}

output:

```

* * * * *
 * * * *
  * * *
   * *
    *

```


6. Half Diamond

```
#include <stdio.h>
```

```
int main() {
```

```
    int i, j;
```

```
    for(i=0; i<5; i++) {
```

```
        for(j=0; j<i; j++) {
```

```
            printf(" * ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    for(i=4; i>0; i--) {
```

```
        for(j=0; j<i; j++) {
```

```
            printf(" * ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

output-

*				
* *				
* * *				
* * * *				
* * * * *				
* * * *				
* * *				
* * *				
* *				
*				

7. Diamond

include <stdio.h>

int main() {

int i, j, k;

for(i=0; i<5; i++) {

for(j=0; j<5-i; j++) {

printf(" ");

}

for(k=0; k<2*i-1; k++) {

printf("* ");

}

printf("\n");

}

for(i=5; i>0; i--) {

for(j=0; j<5-i; j++) {

printf(" ");

}

for(k=0; k<2*i-1; k++) {

printf("* ");

}

printf("\n");

}

return 0;

}

output -

```

      *
    * * *
  * * * * *
* * * * * *
* * * * * *
  * * * * *
    * * *
      *
  
```


5. Pyramid

```
#include <iostream>
using namespace std;
int main() {
```

```
    int i, j, k;
    for(i=0; i<5; i++) {
        for(j=0; j<5-i; j++)
            cout << " ";
```

```
        for(k=0; k<2*i-1; k++)
            cout << " * ";
```

```
        cout << endl;
    }
```

```
    return 0;
}
```

output:

```

      *
     ***
    *****
   *********
  ***********
 *****

```

6. Inverted Pyramid

```
#include namespace std;
using namespace std;
int main() {
```

```
    int i, j, k;
    for(i=5; i>0; i--) {
        for(j=0; j<5-i; j++)
            cout << " ";
```

```
        for(k=0; k<2*i-1; k++)
            cout << " * ";
```

```
        cout << endl;
    }
```

```
    return 0;
}
```

output:

```

*****
 ****
  ***
   **
    *

```


B. C++ Program

1. Square

```
#include <iostream>
using namespace std;
```

```
int main() {
    int i, j;
```

```
    for (i=0; i<5; i++) {
        for (j=0; j<5; j++) {
            cout << "*" ;
        }
        cout << endl;
    }
    return 0;
}
```

output.

*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*

2. Hollow Square

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int i, j;
```

```
    for (i=0; i<5; i++) {
```

```
        for (j=0; j<5; j++) {
```

```
            if (i==0 || i==4 || j==0 || j==4)
```

```
                cout << "*" ;
```

```
            else
```

```
                cout << " " ;
```

```
        }
```

```
        cout << endl;
    }
```

```
}
```

output

*	*	*	*	*
*				*
*				*
*				*
*	*	*	*	*

3. Triangle

```
#include <iostream>
using namespace std;
int main() {
    int i, j;
    for(i=0; i<5; i++) {
        for(j=0; j<5; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

output -

*
* *
* * *
* * * *
* * * * *

4. Inverted Triangle

```
#include <iostream>
using namespace std;
int main() {
    int i, j;
    for(i=5; i>0; i--) {
        for(j=0; j<i; j++) {
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

output:

* * * * *
* * * *
* * *
* * *
* *
*

5. Pyramid

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int i, j, k;
```

```
    for(i=0; i<5; i++) {
```

```
        for(j=0; j<5-i; j++)
```

```
            cout << " ";
```

```
        for(k=0; k<2*i-1; k++)
```

```
            cout << "*";
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

output:

```

      *
     ***
    *****
   *********
  ***********
 
```

6. Inverted Pyramid

```
#include namespace std;
```

```
using namespace std;
```

```
int main() {
```

```
    int i, j, k;
```

```
    for(i=5; i>0; i--) {
```

```
        for(j=0; j<5-i; j++)
```

```
            cout << " ";
```

```
        for(k=0; k<2*i-1; k++)
```

```
            cout << "*";
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

output:

```

*****
 ***
  *
 
```


7. Half Diamond

```
#include <iostream>
using namespace std;
int main() {
    int i, j;

    for(i=0; i<5; i++){
        for(j=0; j<5; j++){
            cout << " * ";

            cout << endl;
        }

        for(i=4; i>0; i--){
            for(j=0; j<i; j++){
                cout << " * ";

                cout << endl;
            }

            return 0;
        }
    }
```

output -

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
.
```


8. Diamond

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int i, j, k;
```

```
    for(i=0; i<5; i++) {
```

```
        for(j=0; j<5-i; j++)
```

```
            cout << " ";
```

```
        for(k=0; k<2*i-1; k++)
```

```
            cout << " * ";
```

```
        cout << endl;
```

```
    }
```

```
    for(i=5; i>0; i--) {
```

```
        for(j=0; j<5-i; j++)
```

```
            cout << " ";
```

```
        for(k=0; k<2*i-1; k++)
```

```
            cout << " * ";
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

output:

```

      *
    * * *
  * * * * *
* * * * * *
* * * * * * *
* * * * * * *
  * * * * *
    * * *
      *
  
```

Programs Execution and Outputs

C Program

[illegible]

Programiz

C Online Compiler

```
1 #include<stdio.h>
2
3 int main() {
4
5     int i, k;
6
7     for(i=5; i>0; i--)
8     {
9         for(k=0; k<i; k++ )
10        {
11            printf("*");
12        }
13        printf("\n");
14    }
15
16    return 0;
17 }
18
19
```

Run

Output

**
*
=== Code Execution Successful ===

JS

TS

GO

[illegible]

[illegible]

[illegible]

C++ Programs

main.cpp

Run

1 #include <iostream>

2 using namespace std;

3

4 int main() {

5 int i, j;

6

7 for(i = 0; i < 5; i++) {

8 for(j = 0; j < 5; j++) {

9 if(i == 0 || i == 4 || j == 0 || j == 4)

10 cout << "*";

11 else

12 cout << " ";

13 }

14 cout << endl;

15 }

16

17 return 0;

18 }

Output

Clear

* *

* *

* *

=== Code Execution Successful ===

JS

TS

main.cpp

Run

Output

Clear

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int i, j;
6
7     for(i = 5; i > 0; i--) {
8         for(j = 0; j < i; j++) {
9             cout << "*";
10        }
11        cout << endl;
12    }
13
14    return 0;
15 }
```

```
*****
****
***
**
*

=== Code Execution Successful ===
```



```
main.cpp  [ ] [ ] [ ] [Run] [Output] [Clear]

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int i, j, k;
6
7      for(i = 4; i >= 0; i--) {
8
9          for(j = 0; j < 4 - i; j++)
10             cout << " ";
11
12         for(k = 0; k < 2 * i + 1; k++)
13             cout << "**";
14
15         cout << endl;
16     }
17
18     return 0;
19 }
```


*

=== Code Execution Successful ===

main.cpp

Run

Output

Clear

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int i, j;
6
7     for(i = 0; i < 5; i++) {
8         for(j = 0; j <= i; j++)
9             cout << "*";
10        cout << endl;
11    }
12
13    for(i = 3; i >= 0; i--) {
14        for(j = 0; j <= i; j++)
15            cout << "*";
16        cout << endl;
17    }
18
19    return 0;
20 }
```

```
*
**
***
****
*****
****
***
**
*

=== Code Execution Successful ===
```



```
main.cpp  Run  Output  Clear
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int i, j, k;
6
7     for(i = 0; i < 5; i++) {
8
9         for(j = 0; j < 4 - i; j++)
10             cout << " ";
11
12         for(k = 0; k < 2 * i + 1; k++)
13             cout << "*";
14
15         cout << endl;
16     }
17
18     for(i = 3; i >= 0; i--) {
19
20         for(j = 0; j < 4 - i; j++)
21             cout << " ";
22
23         for(k = 0; k < 2 * i + 1; k++)
24             cout << "*";
25
26         cout << endl;
27     }
28
29     return 0;
30 }
```

```
*
***
*****
*****
*****
*****
***
*

=== Code Execution Successful ===
```

